



**TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
THAPATHALI CAMPUS**

**A MAJOR PROJECT FINAL REPORT
ON**

**"EVALUATING THE EFFECTIVENES OF VISVAP-PROGRAMMED
SEMI-ACTUATED SIGNAL CONTROL OVER EXISTING FIXED-TIME CONTROL:
A CASE STUDY OF BABARMAHAL INTERSECTION, KATHMANDU VALLEY "**

Submitted By:

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Bidur Ghimire (THA078BCE028)
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Submitted To:

Department of Civil Engineering
Thapathali Campus
Kathmandu, Nepal

Under the Supervision of

Dr. Rojee Pradhananga

February 2025



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ABSTRACT

150-300 words, include: Problem context, Approach, Key results, Impact. Avoid citations, unexpanded acronyms, detailed implementation.

Keywords—Combinatorial optimization, constraint satisfaction, genetic algorithm, hyper-heuristics, metaheuristic search, NP-hard problems, reinforcement learning, university course timetabling problem.

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1. INTRODUCTION

CHAPTER GUIDE: Purpose: set context, define the problem, and state objectives + scope. Must include: Background, Problem statement, Objectives, Scope and limitations. How to write: mention domain + current approaches; quantify gaps with metrics. LaTeX: Cite with cite key, reference figures and tables. General rules: formal/technical style; cite non-trivial claims; reference figures/tables in text.

1.1. Background

This project uses Genetic Algorithm (GA) and Machine Learning (ML) techniques [1]. We also implement Reinforcement Learning (RL) with Application Programming Interface (API) integration. The Central Processing Unit (CPU) and Graphics Processing Unit (GPU) work together for processing.

Research shows optimization methods are crucial in modern computing [2]. Following IEEE guidelines [3,4], we implement state-of-the-art algorithms.

The learning rate α is crucial, and we use discount factor γ . Standard deviation σ helps measure variance with mean μ .

1.2. Problem Statement

1.3. Objectives

Objectives should be SMART: Specific, Measurable, Achievable, Relevant, Time-bound. List primary and secondary objectives clearly.

1.3.1. Scope of the Study

1.4. Research Questions

1.5. Limitations

2. LITERATURE REVIEW

3. THEORETICAL BACKGROUND

3.1. Terminologies and Basic Definitions

3.2. Traffic Flow

3.2.1. Types of Traffic Flow Pattern

3.2.2. Traffic Conflicts

3.3. Level of Service(LOS)

3.4. Capacity

3.5. Vehicle to Capacity Ratio

3.6. Headway

3.6.1. Time Headway

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3.7. Saturation Flow

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3.9. Peak Hour Factor(PHF)

3.10. Unsignalized Intersection

3.11. Signalized Intersection

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4. METHODOLOGY

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5.3. Volume Data in flow Diagram

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6.1. Traffic Flow Calculation

6.2. Calculation of Saturation Flow

6.3. Calculation of Percentage of Turning Movement

6.4. Calculatin of Percentage of Heavy Vehicles

6.5. Computation of Peak Hour Factor(PHF) and Adjusted Volume

6.6. Calculatin of LOS Using V/C Ratio

6.7. Improvement Analysis

6.7.1. Signal Optimization

6.7.2. Comparision of Intersection Parameters

7. RESULTS AND DISCUSSION

8. CONCLUSION AND RECOMMENDATION

8.1. Conclusion

8.2. Recommendation

A. ANNEX I

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